

SketchMind: Understanding Abstract Sketches with MLLMs for Fine-Grained Sketch-Based Image Retrieval

Changxing Li

School of Artificial Intelligence and
Computer Science, Jiangnan
University
Wuxi, Jiangsu, China
li2changxing@hotmail.com

Donglin Zhang*

School of Artificial Intelligence and
Computer Science, Jiangnan
University
Wuxi, Jiangsu, China
dlinzzhang@gmail.com

Zhikai Hu

Department of Computer Science,
Hong Kong Baptist University
Hong Kong SAR, China
cszkhu@comp.hkbu.edu.hk

Xiao-Jun Wu

School of Artificial Intelligence and
Computer Science, Jiangnan
University
Wuxi, Jiangsu, China
wu_xiaojun@jiangnan.edu.cn

Josef Kittler

Centre for Vision, Speech and Signal
Processing (CVSSP), University of
Surrey
Guildford, Surrey, UK
j.kittler@surrey.ac.uk

Abstract

Fine-Grained Sketch-Based Image Retrieval (FG-SBIR) aims to retrieve images that accurately correspond to abstract hand-drawn sketches, requiring the model to understand sparse and abstract visual cues. Existing methods tend to rely on convolutional networks or metric learning to align sketch and image features, often overlooking the inherent abstraction and semantic ambiguity present in sketches. This limitation results in an insufficient understanding of fine-grained visual details. To address this challenge, we propose SketchMind, a novel method that leverages Multi-modal Large Language Models (MLLMs) to enhance abstract sketch understanding in FG-SBIR. Specifically, we use MLLMs to generate auxiliary textual descriptions based on the given sketches via a Visual Question Answering (VQA) strategy. To effectively incorporate these descriptions, we construct a graph structure with the sketch as the central node and the generated texts as peripheral nodes. A graph attention scheme is employed to perform uncertainty-aware feature fusion, enabling the model to suppress noisy or irrelevant textual information. Furthermore, to enhance both inter- and intra-modal fine-grained alignment, we design a Multi-scale Cross-modal Jigsaw Matching module in combination with a self-supervised learning strategy, which captures local and global visual correspondences across modalities more effectively. Extensive experiments on three benchmark FG-SBIR datasets demonstrate that SketchMind achieves superior performance over existing state-of-the-art methods, proving its effectiveness. Code is available at https://github.com/li1changxing/MLLM_FG_SBIR/.

CCS Concepts

• Information systems → Retrieval tasks and goals.

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License.
WWW '26, Dubai, United Arab Emirates
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792705>

Keywords

Cross-modal retrieval, Hand-drawn sketch, MLLM

ACM Reference Format:

Changxing Li, Donglin Zhang, Zhikai Hu, Xiao-Jun Wu, and Josef Kittler. 2026. SketchMind: Understanding Abstract Sketches with MLLMs for Fine-Grained Sketch-Based Image Retrieval. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792705>

1 Introduction

A hand-drawn sketch provides a concise visual abstraction of the target object, capturing essential structural cues without relying on color or texture. These properties have attracted considerable attention, as sketches offer a convenient and effective means for image retrieval, particularly in scenarios where visual content is complex or difficult to describe in words. Building on these properties, Fine-Grained Sketch-Based Image Retrieval (FG-SBIR) has emerged as a promising retrieval paradigm that uses sketches as queries to retrieve corresponding images, requiring precise instance-level matching across heterogeneous modalities. However, the huge modality gap between hand-drawn sketches and natural images presents significant challenges in establishing cross-modal correspondence [49, 50, 52]. To mitigate this issue, existing methods [9, 35] often utilize metric learning schemes, such as triplet loss [33], to align sketch and image representations in high-dimensional spaces. However, the inherent abstraction and sparsity of sketches limit their ability to capture fine-grained visual information. Moreover, variations in the drawing style introduce inconsistencies in the information conveyed.

To enhance the visual information carried by sketches, recent works [11, 54] incorporate auxiliary textual information to enhance sketch representations, as shown in Figure 1 (a). For example, Sketch2Word [20] transforms sketches into pseudo-text and fuses it with human-written language descriptions. While these methods can introduce additional semantic cues absent in sketches, they often require considerable manual effort in designing and annotating textual queries. In addition, to enhance the sketch modality,

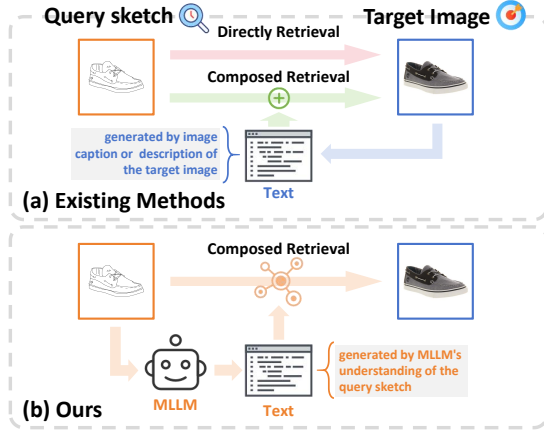


Figure 1: The difference between our method and existing methods. Previous methods use sketches or combine sketches with text extracted from reference images. Our method generates textual descriptions directly from the sketch itself.

some methods [21, 37] use generative models to synthesize photo-realistic images from sketches. For example, Pic2Word [37] maps a sketch to latent space of a pre-trained StyleGAN [16] to generate a image. DSCDM [21] employs sketches as conditional inputs to guide image generation with Stable Diffusion (SD) [31]. However, generative models often introduce visual content that diverges from the intended structure of the input sketch. As illustrated in Figure 2 (Left), using Stable Diffusion (SD) to enhance sketches without any additional (e.g., textual descriptions guidance) can result in images that exhibit structural inconsistencies with the original sketch. For example, the first sketch is transformed into a chair with a fully enclosed backrest, which is absent in the generated sketch. This is likely due to the generative model relying on its learned prior distribution, which may introduce irrelevant visual information. Consequently, these augmented images can hinder rather than help in FG-SBIR tasks due to misaligned semantics.

On the other hand, Multi-modal Large Language Models (MLLMs) have recently demonstrated impressive performance in semantic understanding and cross-modal reasoning. However, their potential in understanding abstract sketches remains underexplored. This observation motivates us to investigate how MLLMs can be leveraged to decode implicit visual information embedded in sketches. In general, existing approaches typically utilize MLLMs for captioning tasks [17, 44], generating holistic descriptions of visual inputs. As illustrated in Figure 2 (Right), such holistic captions are often too coarse to reflect the fine-grained information critical for accurate retrieval. Therefore, how to use MLLMs effectively to understand abstract sketches and extract fine-grained semantic cues remains an open problem that needs further investigation.

To mitigate the above limitations, we propose a novel framework named SketchMind, which leverages MLLMs to generate textual descriptions for sketch augmentation. The proposed SketchMind consists of two main components: i) MLLM-based Sketch

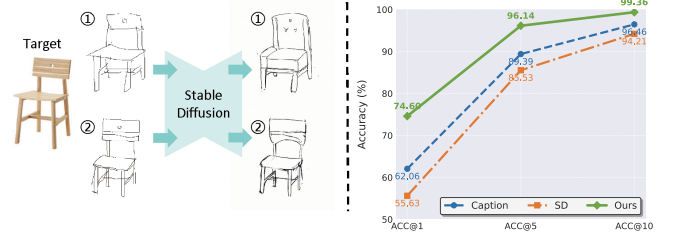


Figure 2: Left: Examples of SD-augmented sketches. Right: The results of Caption (using captions of sketches), SD (using SD-generated sketches) and our approach on QMUL-Chair-V2.

Augmentation (MSA) module and ii) Multi-scale Cross-modal Jigsaw Matching (MCJM) module. In the proposed MSA, we leverage MLLMs to generate textual descriptions for sketches without requiring additional manual annotations. Specifically, we adopt a Visual Question Answering (VQA) strategy to ask targeted questions about object attributes, guiding MLLM to decode the abstract information conveyed by the sketch and generate attribute-based textual descriptions. A graph is then constructed where the sketch feature serves as the central node, and the generated attribute features form the child nodes. Moreover, we utilize a Graph Attention Network (GAT) [41] to dynamically integrate relevant attribute information into the sketch feature while suppressing noisy or irrelevant text features. Meanwhile, the proposed MCJM focuses on cross-modal fine-grained alignment. We divide sketches and images into multi-scale regions and perform contrastive learning on the corresponding parts. In addition, we reconstruct image regions using aligned sketch windows, enhancing the local structural correspondence across modalities.

In summary, our contributions are as follows:

- We propose a novel FG-SBIR framework that integrates MLLMs into abstract sketch understanding via VQA, using the generated descriptions for sketch augmentation.
- We introduce a graph-based fusion scheme, leveraging GAT to dynamically enhance sketch features with attribute-level semantics.
- We develop a MCJM module for fine-grained regional alignment between sketches and images.
- Extensive experiments validate the effectiveness of SketchMind across several FG-SBIR benchmark datasets.

2 Related Work

2.1 Fine-Grained Sketch-based Image Retrieval

Different from category-level SBIR [13, 22, 43], fine-grained SBIR (FG-SBIR) focuses on retrieving instance-level images using sketches. The main challenge lies in aligning fine-grained cues between abstract sketches and the corresponding images. To solve this challenge, early FG-SBIR works [8, 47] usually leverage triplet loss [47] to learn discriminative feature representations. Subsequent approaches enhance this alignment by introducing attention schemes [38]. To reduce the impact of noisy strokes on retrieval performance, [8] employs reinforcement learning for stroke selection,

and [9] treats sketches as stroke sequences and models their temporal structure. To capture fine-grained information, [7] utilizes unlabeled images to enrich the latent semantic representation. Considering the inherent lack of detail in sketches, [18] uses StyleGAN to generate photorealistic images, though this process may introduce irrelevant information. To enable more stable augmentation, Sketch2Word [20] maps sketches into a textual space to generate pseudo-text, which is then used for retrieval with auxiliary text. The above methods can achieve good performance, but most of them rely on leveraging information from images to enhance sketch representations. However, such reliance may overlook the intrinsic potential of the sketches themselves. Therefore, how to extract rich and discriminative information from abstract sketches remains an open and important challenge.

2.2 Handling Abstraction of Sketch

Compared with natural images, hand-drawn sketches are inherently abstract [21, 45], with their level of detail varying depending on the drawing skill, style, and intent. Early work defines sketch abstraction as a function of drawing time [6], where shorter durations correspond to higher levels of abstraction. Later, other methods [27, 28] quantify the level of abstraction based on the stroke count. [3] decodes sketches at varying levels of detail by controlling the number of strokes. In addition, [4] analyzes sketches at coarse and partial stroke levels through gradient backpropagation. More recently, [5] learns sketch abstraction in a semi-supervised manner using codebook learning, while AbstractAway [19] models the abstraction in the smooth latent space of a pre-trained StyleGAN. The above methods rely on implicit modeling or stroke manipulation, which may not capture the abstract semantic information of sketches comprehensively.

2.3 MLLMs for Retrieval

Recently, the emergence of MLLMs [1, 25] has enhanced visual-text modeling and introduced a new paradigm for cross-modal retrieval [42, 51, 53]. By leveraging rich pre-trained world knowledge, MLLMs can generate textual descriptions for the given images or videos, reducing the semantic gap between modalities. For example, Cap4Video [44] demonstrates improved retrieval performance by generating informative captions for the video content. To improve the quality of generated captions so as to enhance retrieval performance, some methods incorporate reasoning strategies such as chain-of-thought (CoT) reasoning for compositional retrieval [39], VQA-based textual enhancement for person re-identification [23], and GPT-4V-guided text generation for richer image descriptions [14]. However, these methods focus primarily on natural images or videos with rich textures and contexts, making it unclear how to adapt MLLMs to abstract inputs like sketches, which lack such cues and pose challenges for reliable text generation.

3 Proposed Method

3.1 Cross-Modal Feature Learning

Given a candidate image set $\mathcal{I}$, FG-SBIR aims to retrieve the corresponding image based on a sketch query $s_q \in \mathcal{S}$, where $\mathcal{S}$ is the sketch set. To this end, a cross-modal joint embedding space $\phi : \mathcal{S} \cup \mathcal{I} \rightarrow \mathbb{R}^d$ is constructed, where the goal is to ensure

that the cosine similarity between a sketch and its corresponding image is higher than that of mismatched pairs. As shown in Figure 3, our proposed SketchMind adopts modality-specific encoders $\mathcal{F}^s$ and $\mathcal{F}^v$ based on ViT [12], which extract sketch and image embeddings as $F^s = \{f^s, R^s\} = \{f^s, r_1^s, \dots, r_{196}^s\}$ and $F^v = \{f^v, R^v\} = \{f^v, r_1^v, \dots, r_{196}^v\}$ respectively, where f^s and f^v denote global features. R^s and R^v represent local features. In addition, to enhance the robustness of intra-modal representation, data augmentation techniques such as rotation and scaling are used on both sketches and images, yielding augmented features $\tilde{f}^s$ and $\tilde{f}^v$. We then apply contrastive learning to enforce intra-modal consistency, formulated as:

$$\mathcal{L}_{self}^s = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(f_i^s \cdot \tilde{f}_i^s / \tau)}{\sum_{j=1}^N \exp(f_i^s \cdot f_j^s / \tau)}, \quad (1)$$

$$\mathcal{L}_{self}^v = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(f_i^v \cdot \tilde{f}_i^v / \tau)}{\sum_{j=1}^N \exp(f_i^v \cdot f_j^v / \tau)}, \quad (2)$$

where τ is a temperature parameter, and N is the batch size. Through intra-modal self-supervised learning, more robust features can be obtained, mitigating the impact of variations in scale or angle.

3.2 MLLM-based Sketch Augmentation

The abstract and sparse nature of hand-drawn sketches poses a key challenge for FG-SBIR. Specifically, sketch encoders often struggle with the ambiguity and lack of structure in sketches. To mitigate this, we develop a sketch enhancement scheme that employs MLLMs to generate semantically rich text, providing auxiliary cues to strengthen the sketch representation. The sketch augmentation module consists of three stages: visual attributes extraction, auxiliary descriptions construction and graph-based information aggregation. We detail these components in the following subsections.

3.2.1 Visual Attributes Extraction. Considering the effectiveness of MLLMs, we leverage LLaVA-NeXT $\mathcal{F}^M$ [26] to capture fine-grained semantics for abstract sketch understanding. Specifically, for the FG-SBIR task, accurate retrieval relies on fine-grained visual information, which is often hard to pinpoint in sketches. To solve this problem, we adopt a visual question answering (VQA) scheme by designing questions that query specific object attributes. Taking shoe retrieval as an example, we leverage GPT-4o to construct an attribute set $A = \{kind, toe\ shape, heel\ height, \dots\}$ covering 7 sketch-relevant aspects ($|A| = 7$) that align with the nature of sketch inputs. Based on A , we construct a corresponding question set Q and leverage MLLM ($\mathcal{F}^M$) to output attribute information. Given a sketch s , the attribute can be obtained as:

$$\tilde{A} = \mathcal{F}^M(s \circ Q), \quad (3)$$

where the input is formed by concatenating the sketch s and the corresponding questions Q . Through this process, $\mathcal{F}^M$ generates semantic attributes $\tilde{A}$ that reflect the underlying semantic information of the sketch. Moreover, as certain attributes may be visually unclear or not depicted, we allow the model to return None for unidentifiable or irrelevant attributes to avoid introducing noise. More details about attributes and question construction are provided in the supplementary material.

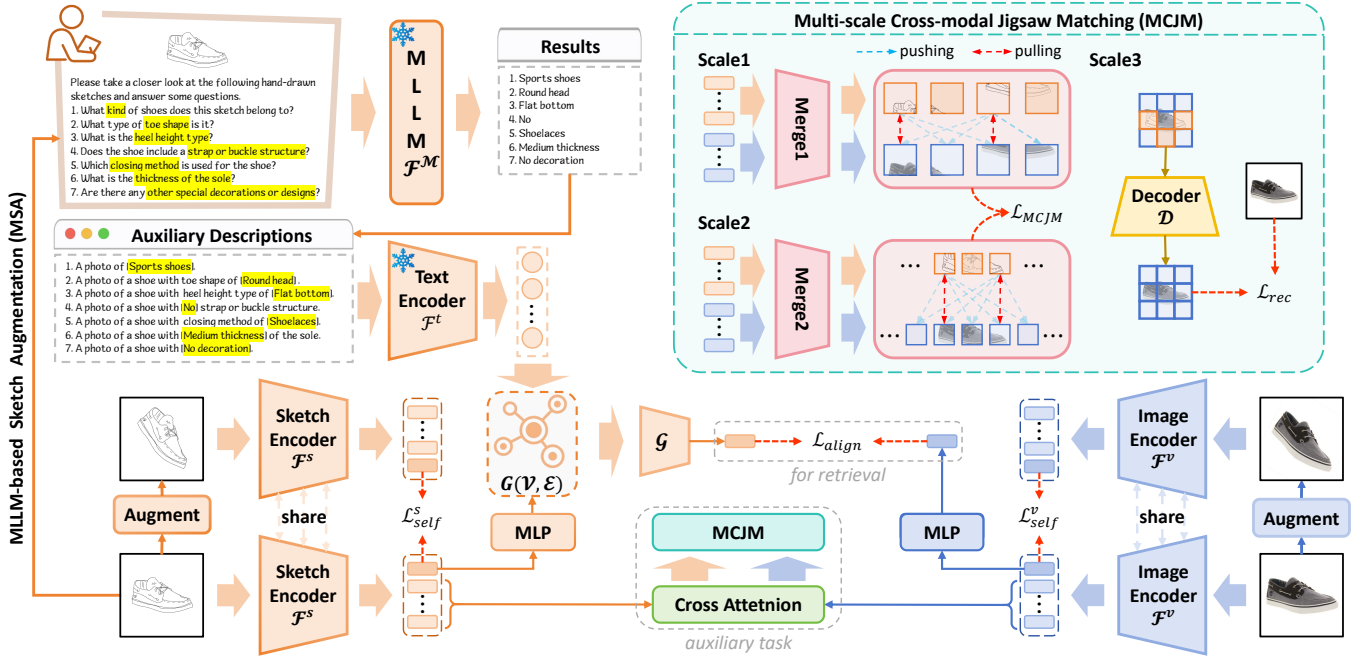


Figure 3: An overview of our proposed SketchMind. Our model includes two main components, *i.e.*, MLLM-based Sketch Augmentation (MSA) and Multi-scale Cross-modal Jigsaw Matching (MCJM). MSA generates attribute descriptions via VQA and enhances sketch features using a graph attention network $\mathcal{G}$. MCJM divides sketches and images into multi-scale windows and performs contrastive learning and reconstruction on corresponding regions.

3.2.2 The Construction of Auxiliary Descriptions. In the second stage, we construct an auxiliary textual description set $T = \bigcup_{i=1}^{|A|} t_i$ based on the generated $\bar{A}$. Specifically, each attribute is converted into a CLIP-compatible sentence t_i using the template A photo of a shoe with <attribute type> of [value]. As the template is not suitable for all attributes, we adjust it for specific cases, as shown in Figure 3 (we provide additional details in the supplementary material). We then use the CLIP [30] textual encoder $\mathcal{F}^t$ to obtain feature embeddings from T , formulated as:

$$F^t = \bigcup_{i=1}^{|A|} \mathcal{F}^t(t_i) = \bigcup_{i=1}^{|A|} f_i^t, \quad (4)$$

where f_i^t denotes the feature of sentence t_i . These features F^t are subsequently used to augment the sketch representation.

3.2.3 Graph-based Information Aggregation. The third step aggregates the learned text features F^t into the sketch representations. Specifically, to capture the one-to-many relationship between a sketch and its associated attribute cues, we employ a graph-based fusion scheme. The sketch is treated as a central node connected to multiple attribute nodes, reflecting the "one sketch, multiple attributes" structure. Formally, given sketch feature f^s and text feature F^t , we construct a graph $G(\mathcal{V}, \mathcal{E})$ to perform information fusion, where $\mathcal{V}$ and $\mathcal{E}$ denote the node set and edges, respectively. Let $\mathcal{V} = \bigcup_{i=0}^{|A|} v_i$, with $v_0 = \delta^s(f^s)$, $v_i = f_i^t$ ($i = 1, \dots, |A|$), where $\delta^s(\cdot)$ is an MLP projection. Edges are defined as $\mathcal{E} = \{(v_i, v_0) | i \in H\}$, where $H = \{i | A_i \neq \text{None}\}$ denotes the indices of available attributes.

Moreover, a GAT layer $\mathcal{G}$ is then used to adaptively aggregate information from the attribute nodes into the sketch node. The fused sketch feature h^s can be computed as:

$$h^s = \sigma\left(\sum_{i \in H} \alpha_{0,i} \mathbf{W} v_i\right), \quad (5)$$

where $\sigma(\cdot)$ is a nonlinear activation, $\mathbf{W}$ is a learnable weight parameter. The attention score $\alpha_{0,i}$ can be formulated as:

$$\alpha_{0,i} = \frac{\exp(\text{LeakyRelu}(a^\top [\mathbf{W} v_0 || \mathbf{W} v_i]))}{\sum_{j \in H} \exp(\text{LeakyRelu}(a^\top [\mathbf{W} v_0 || \mathbf{W} v_j]))}, \quad (6)$$

where a is learnable parameter and $||$ denotes concatenation.

Through the graph attention mechanism, h^s captures both the visual content of the sketch and its associated textual attributes, yielding a robust representation for FG-SBIR. In addition, to align different modalities, we perform contrastive learning on h^s and h^v , where $h^v = \delta^v(f^v)$. The loss function is defined as:

$$\mathcal{L}_{align} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(h_i^s \cdot h_i^v / \tau)}{\sum_{j=1}^N \exp(h_i^s \cdot h_j^v / \tau)}. \quad (7)$$

In the test phase, cosine similarity between h^s and h^v is used for ranking.

3.3 Multi-scale Cross-modal Jigsaw Matching

Beyond the global alignment of sketches and images, capturing fine-grained cross-modal correspondence is essential for FG-SBIR. To this end, we propose a Multi-scale Cross-modal Jigsaw Matching (MCJM) module to learn local-level associations. Specifically, given

that sketches are often abstract, incomplete, and spatially inconsistent with images, we construct jigsaw windows at two spatial scales for both modalities. To establish local correspondences between R^s and R^v , we first apply a cross-attention module to obtain fused visual features $R_F^s, R_F^v \in \mathbb{R}^{14 \times 14 \times d}$. We then apply a convolutional layer ω_1 to reconstruct feature map R_F^s and R_F^v into 2×2 non-overlapping larger windows $P_l^s = \{p_{l,i}^s\}_{i=1}^4$ and $P_l^v = \{p_{l,i}^v\}_{i=1}^4$, with each window containing 7×7 visual tokens. Similarly, another convolutional layer ω_2 generates 7×7 smaller windows $P_s^s = \{p_{s,i}^s\}_{i=1}^{49}$ and $P_s^v = \{p_{s,i}^v\}_{i=1}^{49}$ of size of 2×2 . To enforce cross-modal consistency of fine-grained cues, we consider it as a jigsaw matching problem. Specifically, we conduct contrastive learning at two spatial scales, formulated as:

$$\mathcal{L}_{MCJM} = -\frac{\lambda_l}{|P_l^v|} \sum_{i=1}^{|P_l^v|} \log \frac{\exp(p_{l,i}^s \cdot p_{l,i}^v / \tau)}{\sum_{j=1}^{|P_l^v|} \exp(p_{l,i}^s \cdot p_{l,j}^v / \tau)} - \frac{\lambda_s}{|P_s^v|} \sum_{i=1}^{|P_s^v|} \log \frac{\exp(p_{s,i}^s \cdot p_{s,i}^v / \tau)}{\sum_{j=1}^{|P_s^v|} \exp(p_{s,i}^s \cdot p_{s,j}^v / \tau)}, \quad (8)$$

where λ_l and λ_s are learnable weights for large- and small-scale matches, respectively. The above loss can guide the model to identify fine-grained cross-modal correspondences and mitigate the misalignment introduced by sketch abstraction and sparsity. To enhance the modality consistency, we introduce a jigsaw reconstruction scheme. Specifically, we randomly replace a portion of image tokens R_F^v with sketch tokens R_F^s at corresponding positions, forming a mixed image feature map R_m^v . In addition, we employ a UNet-based [32] decoder $\mathcal{D}$ to reconstruct the original image feature, yielding $\tilde{R}^v = \mathcal{D}(R_m^v)$. The reconstruction is supervised using an MSE loss:

$$\mathcal{L}_{rec} = \frac{1}{|R_F^v|} \sum_{i=1}^{|R_F^v|} \|r_i^v - \tilde{r}_i^v\|, \quad (9)$$

where $\tilde{r}_i^v$ is the output of decoder at position i . The auxiliary loss is defined as $\mathcal{L}_{aux} = \mathcal{L}_{MCJM} + \mathcal{L}_{rec}$. Finally, the overall loss of our method is denoted as:

$$\mathcal{L}_{total} = \mathcal{L}_{align} + \mathcal{L}_{aux} + \beta \mathcal{L}_{self}, \quad (10)$$

where $\mathcal{L}_{self} = \mathcal{L}_{self}^s + \mathcal{L}_{self}^v \cdot \beta$ is a weighting hyperparameter. In this way, our method can enhance the sketch by decoding the abstraction and increasing the alignment of fine-grained cues between the sketch and the image. As a result, the learned representation becomes more discriminative, leading to improved retrieval performance.

4 Experiments

4.1 Experimental Settings

4.1.1 Datasets. We evaluate the proposed SketchMind on three widely used FG-SBIR datasets, Clothes-V1 [15], QMUL-Chair-V2 and QMUL-Shoe-V2 [48]. To be specific, Clothes-V1 contains 1200 (500) sketches (images), with 925 (380) and 275 (120) for training and testing sets. Each image is paired with at least one and up to five sketches. The sketches are drawn by four professional and three junior designers, resulting in relatively high-quality sketches. QMUL-Chair-V2 has 400 images and 1,275 sketches, where each



Figure 4: Comparison of sketches from different datasets.

image is associated with two or three sketches. There are 300 images for training, while 100 for test. QMUL-Shoe-V2 comprises 2,000 shoe images and 6,730 sketches in total. Among them, 5,982 sketches are used for training and 666 for testing. As shown in Figure 4, compared to Clothes-V1, sketches in QMUL-Chair-V2 and QMUL-Shoe-V2 are drawn in a rougher style, making them more abstract and challenging to align with the corresponding images.

4.1.2 Competitors. We compare our approach with recent state-of-the-art methods in FG-SBIR, including: **Triplet-SN** [47] uses triplet loss for FG-SBIR. **Triplet-Att-SN** [38] which introduces an attention module with triplet loss. **B-Siamese** [35] uses a more stronger backbone. **OnTheFly** [9] employs reinforcement learning (RL) with sketch strokes. **Semi-Sup** [7] introduces semi-supervised framework in FG-SBIR. **StyleMeUP** [36] design meta-learning framework for unseen drawing style. **Part-SBIR** [10] studies domain associations via optimal transporting. **NT-SBIR** [8] also employs RL but for noisy stroke removal. **PictureThatSketch** [18] generates photorealistic images using sketches via StyleGan. **Pic2Word** [37], **SEARLE** [2] and **Sketch2Word** [20] introduce auxiliary text description for sketch augmentation via combining sketches and text or mapping sketches into textual space. For that, these three methods can be considered as composed retrieval. **AbstractAway** [19] analyzes abstraction through the generation process. **ARNet** [15] recycles visual features to enhance feature representation.

Table 1: The comparative results of experiments in fine-grained sketch-based image retrieval using the Clothes-V1 dataset.

Method	Acc.@1	Acc.@5	Acc.@10
Triplet-SN	64.36	85.82	92.73
Triplet-Att-SN	70.18	83.64	91.64
B-Siamese	84.73	97.82	<u>99.27</u>
OnTheFly	63.27	90.18	92.73
SketchPVT	89.47	94.73	97.36
ARNet	<u>94.12</u>	<u>98.91</u>	<u>99.27</u>
Ours (SketchMind)	94.55	99.64	99.64

4.1.3 Implementation Details. All experiments are conducted using PyTorch on a single NVIDIA RTX 3090 (24GB) GPU. We adopt the CLIP/16 model as the backbone for both the visual and text encoders, initialized with its pre-trained weights. The MLP heads

Table 2: The comparative results of experiments in fine-grained sketch-based image retrieval using the QMUL-ChairV2 and QMUL-ShoeV2 datasets. “S” means retrieval using sketch only, “S+T” denotes using both sketch and text for composed retrieval. “T” represents model-generated text rather than user-provided input. The best and second best are highlighted in bold and underlined, respectively.

Method	Query	QMUL-Chair-V2			QMUL-Shoe-V2		
		Acc.@1	Acc.@5	Acc.@10	Acc.@1	Acc.@5	Acc.@10
Triplet-SN [47]	S	33.75	65.94	79.26	18.62	43.09	59.31
Triplet-Att-SN [38]	S	37.15	67.80	82.97	22.67	51.20	65.02
B-Siamese[35]	S	40.56	71.83	85.76	20.12	48.95	63.81
OnTheFly [9]	S	39.01	75.85	87.00	35.91	66.78	78.54
SketchAA [46]	S	52.89	-	94.88	32.22	-	79.63
Semi-Sup [7]	S	60.20	78.10	90.81	39.10	69.90	87.50
StyleMeUp [36]	S	62.86	79.60	91.14	36.47	68.10	81.83
Part-SBIR [10]	S	63.30	79.70	-	39.90	68.20	82.90
NT-SBIR [8]	S	64.80	79.10	-	43.70	74.90	-
PictureThatSketch [18]	S	65.10	79.20	-	44.10	75.10	-
SketchPVT [34]	S	71.22	80.10	92.18	44.18	70.80	84.68
Pic2Word [37]	S+T	-	55.70	62.10	-	34.70	58.40
SEARLE [2]	S+T	-	60.80	66.40	-	38.40	64.80
Sketch2Word [20]	S+T	-	73.50	81.40	-	47.30	79.10
AbstractAway [19]	S	72.10	80.90	-	<u>45.30</u>	77.30	-
ARNet [15]	S	<u>73.31</u>	<u>93.24</u>	<u>97.15</u>	40.11	67.54	79.29
Ours (SketchMind)	S+T	74.92	96.14	99.36	45.50	<u>75.53</u>	<u>84.69</u>

δ^s , δ^v contain three fully connected layers with dimensions [512, 1024, 512]. Input images are resized to 224×224 , and the final retrieval feature dimension for both h^s or h^v is set to 512. We set the temperature parameter τ to 0.07, β is 0.5. We train our method using Adam optimizer with learning rate of $6e-6$ and weight decay of $1e-4$ for 300 epochs with batch size of 32. For evaluation, we report Acc.@1, Acc.@5 and Acc.@10, which indicate the percentage of correct matches ranked within the top-K retrieved results ($K = 1, 5, 10$). To improve training efficiency, we employed an offline pre-calculation strategy for sketch attribute generation. All sketches are first processed by the MLLM to obtain their corresponding attribute descriptions, which takes approximately 6 seconds for each sketch.

4.2 Performance Analysis

4.2.1 Comparisons with State-of-the-Art Methods. We report the experimental results obtained on three FG-SBIR benchmarks in Table 1 and Table 2. Our SketchMind consistently outperforms existing methods across all datasets. Specifically, on the Clothes-V1 dataset (with high quality sketch), our method obtains an Acc.@1 of 94.55%, outperforming the previous best (ARNet 94.12%) by 0.43%. Both Acc.@5 and Acc.@10 reach 99.64%, verifies the effectiveness of our SketchMind. On QMUL-Chair-V2 and QMUL-Shoe-V2, where sketches are more abstract and less consistent, our method also achieves promising results. Specifically, on Chair-V2, SketchMind obtains 74.92% on Acc.@1, 96.14% on Acc.@5 and 99.36% on Acc.@10, respectively, outperforming the best-performing method, namely ARNet. On Shoe-V2, our approach sets the highest Acc.@1 (45.50%) and Acc.@10 (84.69%). Moreover, we can see that our

Acc.@5 is slightly lower than that of some competitors. One possible reason is that the model ranks the correct match either very high or slightly lower due to local ranking noise, affecting Acc.@5. Furthermore, compared to the retrieval methods that use “S+T” queries (e.g., Pic2Word), SketchMind achieves substantial gains on Acc.@5, improving by 22.96% on Chair-V2 and 28.23% on Shoe-V2. This improvement can be attributed to the limitations of the caption based text, which often lacks fine-grained detail, and the information loss introduced by projecting sketches into textual space. In addition, SketchMind outperforms the generation-based approach SketchPVT, and our method also surpasses the abstraction-oriented approaches such as AbstractAway, with a 2.82% gain on Chair-V2 in Acc.@1. These results collectively validate the effectiveness of our method.

4.2.2 Ablation Study. To assess the contribution of each component, we conduct some ablation experiments, as shown in Table 3. Specifically, (1) $w/o \mathcal{L}_{self}$ means that the intra-modal self-supervised module is removed. (2) $w/o \mathcal{L}_{MCJM}$ excludes the cross-modal jigsaw matching objective, (3) $w/o \mathcal{L}_{rec}$ indicates that reconstruction module is not used, and (4) $w/o \mathcal{F}^M$ means that MLLM is discarded for abstraction understanding. From these results, we can draw the following observations: (1) Removing $\mathcal{L}_{self}$ leads to a significant drop of 6.3% in Acc.@1, indicating its role in enhancing intra-modal feature discriminability. (2) Excluding $\mathcal{L}_{MCJM}$ results in a 2.68% performance drop, confirming the importance of cross-modal alignment. (3) Without reconstruction module, Acc.@1 decreases by 1.61%, suggesting that reconstruction promotes better modality consistency. (4) Removing $\mathcal{F}^M$ leads to a 3.03% drop,

Table 3: An ablation study of our SketchMind components on QMUL-Chair-V2.

Method	Acc.@1	Acc.@5	Acc.@10
w/o $\mathcal{L}_{self}$	68.33	91.39	95.09
w/o $\mathcal{L}_{MCJM}$	72.24	93.24	96.80
w/o $\mathcal{L}_{rec}$	73.31	95.53	97.15
w/o $\mathcal{F}^M$	71.89	93.24	96.44
FULL	74.92	96.14	99.36

Table 4: Analysis of the fusion module for attribute aggregation on QMUL-Chair-V2.

Method	Acc.@1	Acc.@5	Acc.@10
Pooling	69.04	92.17	96.09
Concatenation	70.46	90.39	96.44
GAT	74.92	96.14	99.36

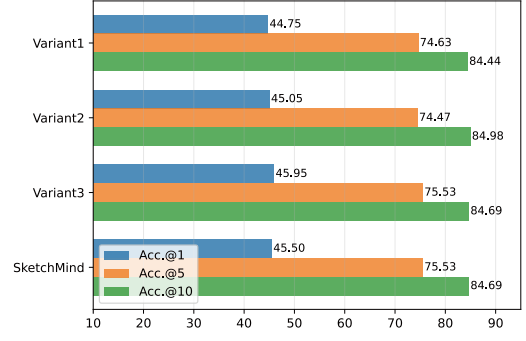
which demonstrates the effectiveness of incorporating MSA module for sketch understanding.

4.3 Additional Analysis

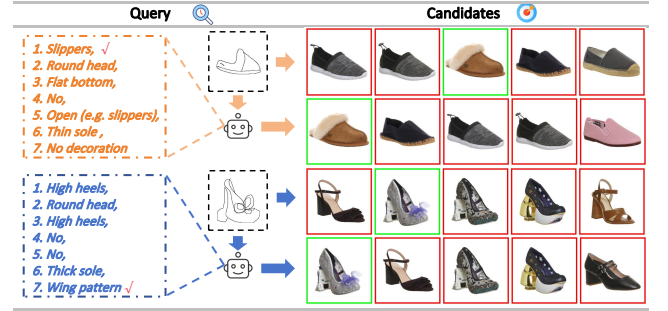
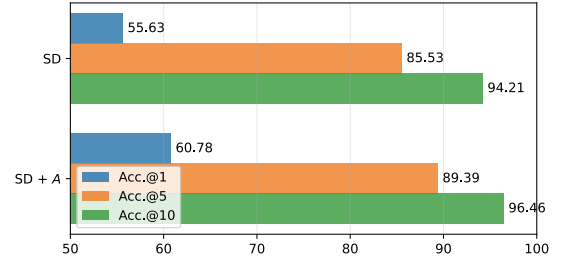
4.3.1 Analysis of GAT Fusion Module. To validate the effectiveness of our GAT for attribute aggregation, we conducted experiment against two simpler fusion strategies in Table 4. Specifically, our GAT module surpasses simple concatenation (70.46%) and average pooling (69.04%) by 4.46% and 5.88%, respectively. This might be because that text generated by MLLM can contain noise or irrelevant information, while pooling and concatenation treat all attributes equally, which can influence the final feature representation with noisy data.

4.3.2 Analysis of Attribute Nodes. To analyze the impact of attributes on the performance, we conduct experiments from three perspectives, as shown in Figure 5. (1) Variant1 (*no A for G*): only the sketch self-loop is used, excluding all attribute nodes. This setting results in a 0.8% drop in Acc.@1, indicating that attributes provide useful information for retrieval. (2) Variant2 (*random select A*): we randomly select 3 to 5 attributes for aggregation. This leads to a 0.45% performance decrease, suggesting that using more attributes can bring more information for sketch augmentation. (3) Variant3 (*A generated from I*): we replace the sketch-derived attributes with those generated from images using MLLMs. We find that the performance of SketchMind is close to that using attributes generated from images, only lagging 0.45% on Acc.@1. This shows that the proposed MLLMs-based scheme can decode abstract sketch cues as effectively as from images.

We also performed a qualitative analysis, as shown in Figure 6. We present a qualitative comparison of the retrieval results using sketch-only and those augmented with MLLM-generated descriptions. As shown, incorporating text generated by the MLLM leads to more accurate matches, with the ground-truth image ranked higher. For example, in the first row, the sketch-only retrieval fails to rank the ground-truth slipper image at Top-1. After incorporating the

**Figure 5: An analysis of the attributes on QMUL-Shoe-V2.**

MLLM-generated attributes, the correct instance is successfully retrieved at the top rank. A similar improvement is observed in the second case. These results illustrate the effectiveness of using MLLM-decoded semantic attributes to guide the FG-SBIR task.

**Figure 6: A comparison of the retrieval results: sketch-only versus sketch with MLLM-generated attributes.****Figure 7: Comparison of FG-SBIR with (SD + A) and without (SD) MLLM-generated attribute guidance on Chair-V2.**

4.3.3 Qualitative Results for the Generative Augmentation Task. Beyond its application in FG-SBIR, our method is also effective for generative sketch augmentation. As illustrated in Figure 8, prompts without explicit attribute constraints (orange) often lead to visually inconsistent generations. For example, in the first case, the generated chair incorrectly includes a full backrest, while in the second, the office chair lacks wheels. By contrast, incorporating fine-grained attribute descriptions via MLLM decoding (green and

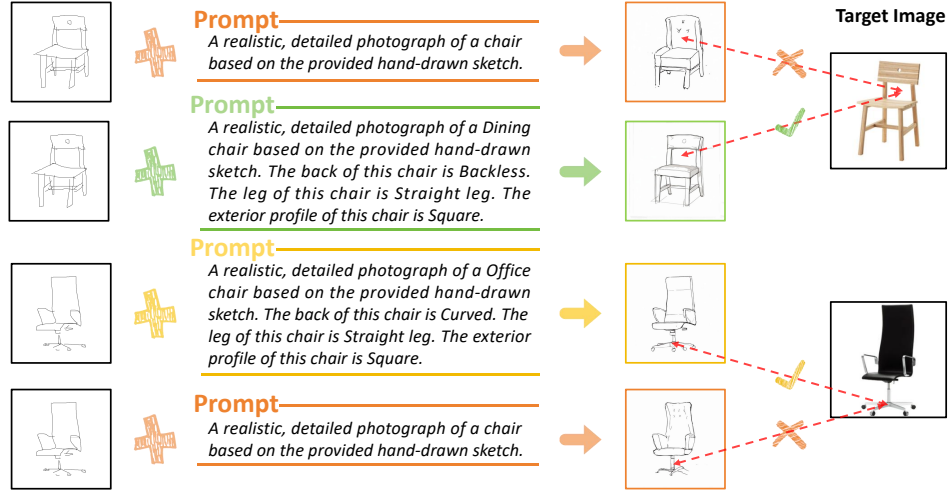


Figure 8: Qualitative results obtained using generative sketch augmentation with pre-trained stable-diffusion model. Orange prompts are without attribute information, while green and yellow prompts incorporate attributes.

yellow prompts) significantly improves generation quality, producing results more faithful to the target image. This suggests that semantic attributes can guide the diffusion process to preserve the structural details present in the sketch. We further evaluate the impact of attribute-enhanced generations in FG-SBIR. As shown in Figure 7, compared to SD-augmented sketches without the attribute guidance, our method yields consistent improvements across three metrics (Acc.@1, Acc.@5 and Acc.@10): +5.15% (60.78%), +3.86% (89.39%), and +2.25% (96.46%). These gains highlight the effectiveness of MLLM-based attribute decoding in capturing abstract sketch semantics and enhancing downstream retrieval tasks.

4.3.4 Zero-shot FG-SBIR. To evaluate the generalization of our method, we follow ZSE [24], by training on QMUL-Shoe-V2 and testing on QMUL-Chair-V2. As shown in Figure 9, our method achieves 95.50% on Acc.@10, surpassing ZSE (94.53%) by 0.97%, and outperforming all other baselines (CC-DG [29], StyleMeUP [36], OnTheFly [9] and Triplet-Att-SN [38]). We attribute this improvement to the use of MLLM to decode abstract visual strokes into explicit text attributes. These attributes can be effectively transferred across categories, confirming the effectiveness of our SketchMind under the zero-shot setting.

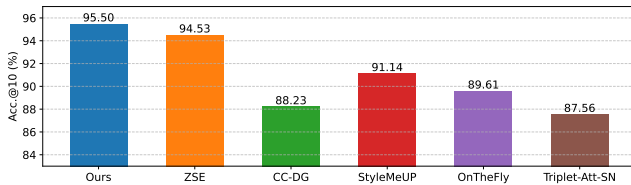


Figure 9: The results of ZS-FG-SBIR on Acc.@10.

4.3.5 Feature Embedding. We visualize feature distribution using t-SNE [40], as shown in Figure 10. Sketches and images form semantically coherent clusters, with each sketch embedding closely

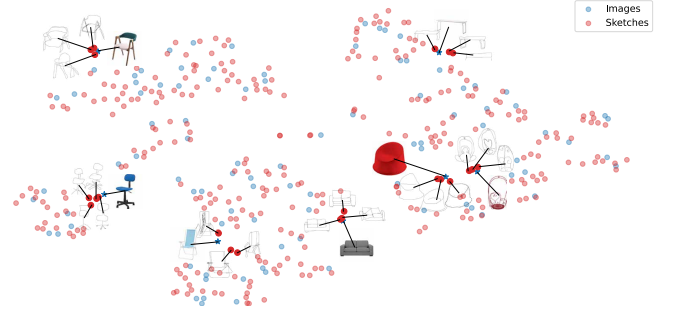


Figure 10: t-SNE visualization of feature embeddings in the joint space on QMUL-Chair-V2.

aligned to its corresponding image. This alignment indicates that our model can capture cross-modal semantics effectively, enabling accurate instance-level retrieval in FG-SBIR.

5 Conclusion

In this paper, we propose a novel method termed SketchMind for abstract sketch understanding in FG-SBIR. The proposed approach first utilizes MLLM to decode sketch cues into attribute descriptions, and then employs a GAT to aggregate these attributes into sketch features. To establish fine-grained alignment between sketches and images, we develop a Multi-scale Cross-modal Jigsaw Matching module combined with a self-supervised learning strategy. Our SketchMind achieves SOTA performance on FG-SBIR, introduces a novel perspective for advancing research, and holds promising potential for future studies on sketch abstraction.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (62202204) and the Fundamental Research Funds for the Central Universities (JUSRP123032).

References

- [1] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. 2023. Qwen technical report. *arXiv preprint arXiv:2309.16609* (2023).
- [2] Alberto Baldrati, Lorenzo Agnolucci, Marco Bertini, and Alberto Del Bimbo. 2023. Zero-shot composed image retrieval with textual inversion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 15338–15347.
- [3] Hmrishav Bandyopadhyay, Ayan Kumar Bhunia, Pinaki Nath Chowdhury, Aneeshan Sain, Tao Xiang, Timothy Hospedales, and Yi-Zhe Song. 2024. Sketchinr: A first look into sketches as implicit neural representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12565–12574.
- [4] Hmrishav Bandyopadhyay, Pinaki Nath Chowdhury, Ayan Kumar Bhunia, Aneeshan Sain, Tao Xiang, and Yi-Zhe Song. 2024. What Sketch Explainability Really Means for Downstream Tasks?. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10997–11008.
- [5] Hmrishav Bandyopadhyay, Pinaki Nath Chowdhury, Aneeshan Sain, Subhadeep Koley, Tao Xiang, Ayan Kumar Bhunia, and Yi-Zhe Song. 2024. Do Generalised Classifiers Really Work on Human Drawn Sketches?. In *European Conference on Computer Vision*. Springer, 217–235.
- [6] Itamar Berger, Ariel Shamir, Moshe Mahler, Elizabeth Carter, and Jessica Hodgins. 2013. Style and abstraction in portrait sketching. *ACM Transactions on Graphics (TOG)* 32, 4 (2013), 1–12.
- [7] Ayan Kumar Bhunia, Pinaki Nath Chowdhury, Aneeshan Sain, Yongxin Yang, Tao Xiang, and Yi-Zhe Song. 2021. More photos are all you need: Semi-supervised learning for fine-grained sketch based image retrieval. In *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*. 4247–4256.
- [8] Ayan Kumar Bhunia, Subhadeep Koley, Abdullah Faiz Ur Rahman Khilji, Aneeshan Sain, Pinaki Nath Chowdhury, Tao Xiang, and Yi-Zhe Song. 2022. Sketching without worrying: Noise-tolerant sketch-based image retrieval. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 999–1008.
- [9] Ayan Kumar Bhunia, Yongxin Yang, Timothy M Hospedales, Tao Xiang, and Yi-Zhe Song. 2020. Sketch less for more: On-the-fly fine-grained sketch-based image retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9779–9788.
- [10] Pinaki Nath Chowdhury, Ayan Kumar Bhunia, Viswanatha Reddy Gajjala, Aneeshan Sain, Tao Xiang, and Yi-Zhe Song. 2022. Partially does it: Towards scene-level fg-sbir with partial input. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2395–2405.
- [11] Pinaki Nath Chowdhury, Ayan Kumar Bhunia, Aneeshan Sain, Subhadeep Koley, Tao Xiang, and Yi-Zhe Song. 2023. Scenetrilogy: On human scene-sketch and its complementarity with photo and text. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10972–10983.
- [12] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [13] Jiale Du, Yang Liu, Xinbo Gao, Jungong Han, and Lei Zhang. 2025. Zero-Shot Sketch-Based Image Retrieval with teacher-guided and student-centered cross-modal bidirectional knowledge distillation. *Pattern Recognition* 164 (2025), 111529.
- [14] Hongfei Ge, Yuanchun Jiang, Jianshan Sun, Kun Yuan, and Yezheng Liu. 2025. LLM-Enhanced Composed Image Retrieval: An Intent Uncertainty-Aware Linguistic-Visual Dual Channel Matching Model. *ACM Transactions on Information Systems* 43, 2 (2025), 1–30.
- [15] Jianan Jiang, Hao Tang, Zhilin Jiang, Weiren Yu, and Di Wu. 2025. ARNet: Self-Supervised FG-SBIR with Unified Sample Feature Alignment and Multi-Scale Token Recycling. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 3985–3993.
- [16] Tero Karras, Samuli Laine, and Timo Aila. 2019. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4401–4410.
- [17] Shyamgopal Karthik, Karsten Roth, Massimiliano Mancini, and Zeynep Akata. 2023. Vision-by-language for training-free compositional image retrieval. *arXiv preprint arXiv:2310.09291* (2023).
- [18] Subhadeep Koley, Ayan Kumar Bhunia, Aneeshan Sain, Pinaki Nath Chowdhury, Tao Xiang, and Yi-Zhe Song. 2023. Picture that sketch: Photorealistic image generation from abstract sketches. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 6850–6861.
- [19] Subhadeep Koley, Ayan Kumar Bhunia, Aneeshan Sain, Pinaki Nath Chowdhury, Tao Xiang, and Yi-Zhe Song. 2024. How to handle sketch-abstraction in sketch-based image retrieval?. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16859–16869.
- [20] Subhadeep Koley, Ayan Kumar Bhunia, Aneeshan Sain, Pinaki Nath Chowdhury, Tao Xiang, and Yi-Zhe Song. 2024. You'll Never Walk Alone: A Sketch and Text Duet for Fine-Grained Image Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16509–16519.
- [21] Subhadeep Koley, Ayan Kumar Bhunia, Deepanshu Sekhri, Aneeshan Sain, Pinaki Nath Chowdhury, Tao Xiang, and Yi-Zhe Song. 2024. It's All About Your Sketch: Democratising Sketch Control in Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7204–7214.
- [22] Chang-Xing Li, Donglin Zhang, Zhikai Hu, and Xiao-Jun Wu. 2025. Modality fused class-proxy with knowledge distillation for zero-shot sketch-based image retrieval. *IEEE Transactions on Circuits and Systems for Video Technology* (2025).
- [23] Rui Li, Tingting Ren, Jie Wen, and Jinxing Li. 2024. Optimal Transport-based Labor-free Text Prompt Modeling for Sketch Re-identification. *Advances in Neural Information Processing Systems* 37 (2024), 119198–119226.
- [24] Fengyin Lin, Mingkang Li, Da Li, Timothy Hospedales, Yi-Zhe Song, and Yong-gang Qi. 2023. Zero-shot everything sketch-based image retrieval, and in explainable style. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 23349–23358.
- [25] Aixun Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Cheng-gang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. 2024. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437* (2024).
- [26] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yunnan Zhang, Sheng Shen, and Yong Jae Lee. 2024. Llava-next: Improved reasoning, ocr, and world knowledge.
- [27] Umar Riaz Muhammad, Yongxin Yang, Timothy M Hospedales, Tao Xiang, and Yi-Zhe Song. 2019. Goal-driven sequential data abstraction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 71–80.
- [28] Umar Riaz Muhammad, Yongxin Yang, Yi-Zhe Song, Tao Xiang, and Timothy M Hospedales. 2018. Learning deep sketch abstraction. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 8014–8023.
- [29] Kaiyue Pang, Ke Li, Yongxin Yang, Honggang Zhang, Timothy M Hospedales, Tao Xiang, and Yi-Zhe Song. 2019. Generalising fine-grained sketch-based image retrieval. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 677–686.
- [30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PMLR, 8748–8763.
- [31] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- [32] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5–9, 2015, proceedings, part III* 18. Springer, 234–241.
- [33] Aneeshan Sain, Ayan Kumar Bhunia, Pinaki Nath Chowdhury, Subhadeep Koley, Tao Xiang, and Yi-Zhe Song. 2023. Clip for all things zero-shot sketch-based image retrieval, fine-grained or not. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2765–2775.
- [34] Aneeshan Sain, Ayan Kumar Bhunia, Subhadeep Koley, Pinaki Nath Chowdhury, Soumitri Chattopadhyay, Tao Xiang, and Yi-Zhe Song. 2023. Exploiting unlabelled photos for stronger fine-grained SBIR. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6873–6883.
- [35] Aneeshan Sain, Ayan Kumar Bhunia, Yongxin Yang, Tao Xiang, and Yi-Zhe Song. 2020. Cross-modal hierarchical modelling for fine-grained sketch based image retrieval. *arXiv preprint arXiv:2007.15103* (2020).
- [36] Aneeshan Sain, Ayan Kumar Bhunia, Yongxin Yang, Tao Xiang, and Yi-Zhe Song. 2021. Stylemeup: Towards style-agnostic sketch-based image retrieval. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8504–8513.
- [37] Kuniaki Saito, Kihyuk Sohn, Xiang Zhang, Chun-Liang Li, Chen-Yu Lee, Kate Saenko, and Tomas Pfister. 2023. Pic2word: Mapping pictures to words for zero-shot composed image retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 19305–19314.
- [38] Jifei Song, Qian Yu, Yi-Zhe Song, Tao Xiang, and Timothy M Hospedales. 2017. Deep spatial-semantic attention for fine-grained sketch-based image retrieval. In *Proceedings of the IEEE international conference on computer vision*. 5551–5560.
- [39] Yuanmin Tang, Xiaoting Qin, Jue Zhang, Jing Yu, Gaopeng Gou, Gang Xiong, Qingwei Ling, Saravan Rajmohan, Dongmei Zhang, and Qi Wu. 2024. Reason-before-Retrieve: One-Stage Reflective Chain-of-Thoughts for Training-Free Zero-Shot Composed Image Retrieval. *arXiv preprint arXiv:2412.11077* (2024).
- [40] Laurens Van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of machine learning research* 9, 11 (2008).
- [41] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, and Yoshua Bengio. 2017. Graph attention networks. *arXiv preprint arXiv:1710.10903* (2017).
- [42] Zhiwen Wang, Donglin Zhang, and Zhikai Hu. 2024. LSECA: local semantic enhancement and cross aggregation for video-text retrieval. *International Journal of Multimedia Information Retrieval* 13, 3 (2024), 30.
- [43] Weijie Wu, Jun Li, Zhijian Wu, and Jianhua Xu. 2025. Zero-Shot Sketch-Based Image Retrieval with Hybrid Information Fusion and Sample Relationship Modeling. In *International Conference on Multimedia Modeling*. Springer, 337–350.
- [44] Wenhao Wu, Haipeng Luo, Bo Fang, Jingdong Wang, and Wanli Ouyang. 2023. Cap4video: What can auxiliary captions do for text-video retrieval?. In *Proceedings*

of the IEEE/CVF conference on computer vision and pattern recognition. 10704–10713.

- [45] Peng Xu, Yongye Huang, Tongtong Yuan, Tao Xiang, Timothy M Hospedales, Yi-Zhe Song, and Liang Wang. 2020. On learning semantic representations for large-scale abstract sketches. *IEEE Transactions on Circuits and Systems for Video Technology* 31, 9 (2020), 3366–3379.
- [46] Lan Yang, Kaiyue Pang, Honggang Zhang, and Yi-Zhe Song. 2021. Sketchaa: Abstract representation for abstract sketches. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 10097–10106.
- [47] Qian Yu, Feng Liu, Yi-Zhe Song, Tao Xiang, Timothy M Hospedales, and Chen-Chang Loy. 2016. Sketch me that shoe. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 799–807.
- [48] Qian Yu, Jifei Song, Yi-Zhe Song, Tao Xiang, and Timothy M Hospedales. 2021. Fine-grained instance-level sketch-based image retrieval. *International Journal of Computer Vision* 129, 2 (2021), 484–500.
- [49] Donglin Zhang, Changxing Li, and Xiao-Jun Wu. 2025. Multi-level Encoding with Hierarchical Alignment for Sketch-Based 3D Shape Retrieval. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1033–1043.
- [50] Donglin Zhang and Xiao-Jun Wu. 2022. Robust and discrete matrix factorization hashing for cross-modal retrieval. *Pattern Recognition* 122 (2022), 108343.
- [51] Donglin Zhang, Xiao-Jun Wu, and Guoqing Chen. 2023. ONION: Online semantic autoencoder hashing for cross-modal retrieval. *ACM Transactions on Intelligent Systems and Technology* 14, 2 (2023), 1–18.
- [52] Donglin Zhang, Xiao-Jun Wu, and Jun Yu. 2021. Discrete bidirectional matrix factorization hashing for zero-shot cross-media retrieval. In *Chinese conference on pattern recognition and computer vision (PRCV)*. Springer, 524–536.
- [53] Donglin Zhang, Xiao-Jun Wu, and Jun Yu. 2021. Label consistent flexible matrix factorization hashing for efficient cross-modal retrieval. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* 17, 3 (2021), 1–18.
- [54] Ran Zuo, Haoxiang Hu, Xiaoming Deng, Cangjun Gao, Zhengming Zhang, Yu-Kun Lai, Cuixia Ma, Yong-Jin Liu, and Hongan Wang. 2024. SceneDiff: generative scene-level image retrieval with text and sketch using diffusion models. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*. 1825–1833.

A Supplementary

A.1 Attributes Selection.

We display the attributes we selected in Table 5. These attributes are carefully chosen via GPT4o based on their domain-specific importance, ensuring they reflect the essential fine-grained cue features.

A.2 Question Design.

We show the text prompt we designed for VQA based on the above attributes for the three fine-grained categories in Figure 11, 12 and 13, respectively. While we encourage MLLM to output the answer by key phrase for text description sentence construction,

A.3 Description Sentence Construction.

We provide the constructed text descriptions for sketch enhancement in Figure 14, while [attributes] in red font are the corresponding attributes generated from VQA. These sentences are carefully designed to align with the latent space distribution of CLIP. By leveraging CLIP’s pretrained text encoder, our descriptions maintain semantic coherence while maximizing the model’s ability to bridge sketch and text representations.

A.4 Project Code

Code is available at
https://github.com/li1changxing/MLLM_FG_SBIR/.

Table 5: Attributes Selection of Each Fine-grained Category.

Category	Attributes
Shoe	1. kind
	2. toe shape
	3. type of heel height
	4. strap or buckle structure
	5. closing method
	6. thickness of the sole
	7. special decorations
Chair	1. types
	2. back
	3. handrail structure
	4. chair legs
	5. exterior profile of the seat
	6. rotate
	7. foldable features
Clothe	1. type
	2. geometric shape of the collar
	3. length of sleeve
	4. pocket structures
	5. style of overall silhouette
	6. closed structure of garment
	7. patterns

Table 6: Comparison using different MLLM backbones.

VLMs	Acc.@1	Acc.@5
LLaVa-Next-8b	74.92	96.14
Llama-3.1-8b-Instruct	74.92	96.28
DeepSeek-V12-Tiny-8b	74.30	97.21
Qwen3-VL-8B-Instruct	74.61	96.59
Qwen3-VL-30B-Instruct	75.85	96.90

A.5 Evaluation on Different MLLMs

We evaluate recent state-of-the-art open-source VLMs to validate robustness. The results are shown in the Table 6 that the larger Qwen3-VL-30B yields only a marginal gain (+0.93% Acc.@1), while the 8B model provides a better balance between performance and parameter efficiency. This can also confirm that performance is primarily driven by our method rather than solely by VLMs capacity.

Text Prompt:
Please take a closer look at the following hand-drawn sketches and answer some questions. Use None for those that cannot be identified.

1. What kind of shoes does this sketch belong to? (Sports shoes, boots, sandals, high heels, slippers, Oxfords, etc.)
2. The toe shape is closest to which of the following type? (round head, pointed head, square head, almond head, fish head, etc.)
3. Which of the following type of heel height? (Flat, low heels, medium heels, high heels)
4. Does the sketch include a strap or buckle structure? (Yes, No)
5. Which closing method is used for the shoes in the sketch? (shoelaces, Velcro, zippers, open [e.g. slippers], buckles, etc.)
6. What is the thickness of the sole? (Thin sole, medium thickness, thick sole [such as platform shoes])
7. Are there any other special decorations or designs? (wing pattern, stitch pattern, no decoration)

Please answer in the following format:

1. Sports shoes
2. Round head
3. Flat bottom
4. Yes
-

Figure 11: Text Prompt for Shoe.

Text Prompt:
Please take a closer look at the following hand-drawn sketches and answer some questions. Use None for those that cannot be identified.

1. What are the main types of chairs in this sketch? (Dining chair, office chair, bar stool, lounge chair, etc.)
2. The geometry of the back of a chair is closest to which of the following types? (Backless, rectangular, circular, curved, etc.)
3. Does the sketch include handrail structure? (Yes/no)
4. What is the shape of the chair legs? (Straight leg, bent leg, cross leg, etc.)
5. What is the exterior profile of the seat? (Square, round, oval, saddle, irregular)
6. Is the chair designed to rotate? (Yes/no)
7. Does the chair have foldable features? (Yes/no)

Please answer in the following format:

1. Dining chair
2. Backless
3. No
4. Bent leg
-

Figure 12: Text Prompt for Chair.



Figure 13: Text Prompt for Clothes.

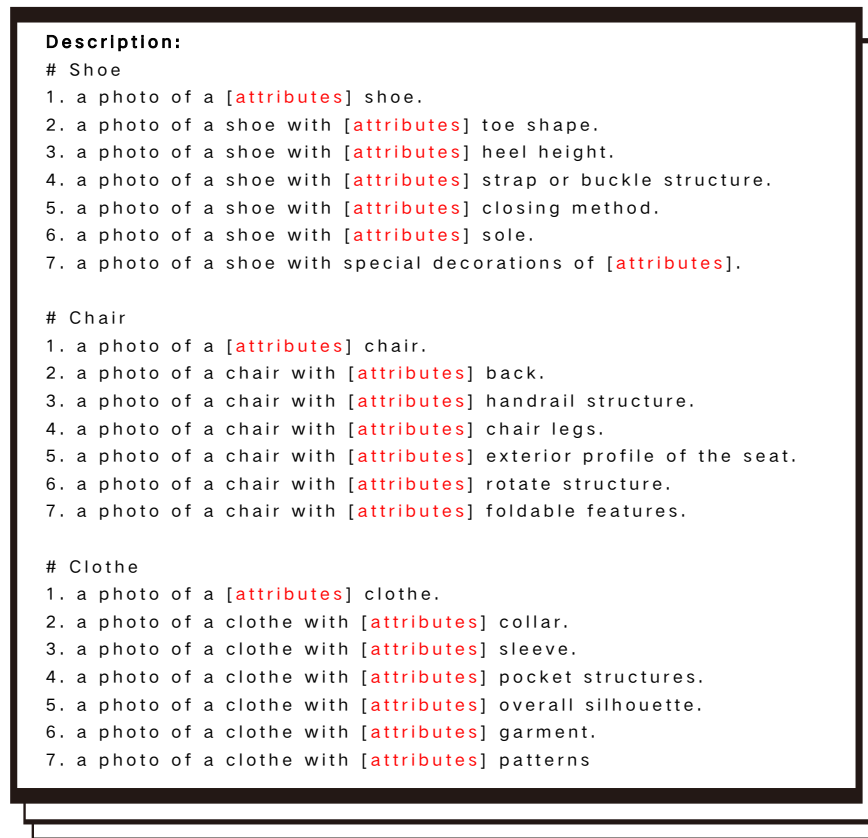


Figure 14: Description Sentence Construction for the three Categories.